Description

Illustration





Figure caption

1. Plywood is described by the number layers or plies as seen from this cross section. 2. Exterior grade plywood used as protection against a hurricane. Images used under license from Shutterstock.com

The material

Plywood is laminated wood, the layers glued together such that the grain in successive layers are at right angles, giving stiffness and strength in both directions. The number of layers varies, but is always odd (3, 5, 7 etc.) to give symmetry about the core ply - if it is unsymmetric it warps when wet or hot. Those with few plies (3,5) are significantly stronger and stiffer in the direction of the outermost layers; with increasing number of plies the properties become more uniform. High quality plywood is bonded with synthetic resin.

Compositional summary

Cellulose/Hemicellulose/Lignin/12%H2O/Adhesive

General properties

Density	700	-	800	kg/m^3
Price	* 0.415	-	0.461	GBP/kg
Date first used	1907			

Material form that data applies to

Bulk	✓
Sheet	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✗

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Mechanical properties

Young's modulus		3.4	-	5.1	GPa
Shear modulus		0.47	-	0.76	GPa
Bulk modulus	*	1.63	-	2.45	GPa
Bending modulus	*	6.9	-	13	GPa
Poisson's ratio		0.2	-	0.3	
Yield strength (elastic limit)		4	-	6	MPa
Tensile strength		4	-	6	MPa
Compressive strength		17.5	-	26.5	MPa
Bending strength		15	-	35	MPa
Elongation	*	2.38	-	2.91	% strain
Hardness - Vickers		5.83	-	8.83	HV
Fatigue strength at 10^7 cycles	*	2.9	-	4.4	MPa
Fracture toughness	*	0.5	-	1	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.012	-	0.017	

Thermal and combustion properties

Thermal conductor or insulator?		Good insulator			
Thermal resistivity		6.67	-	8.33	m.°C/W
Thermal expansion coefficient		6	-	8	µstrain/°C
Specific heat capacity		1.66e3	-	1.71e3	J/kg.°C
Maximum service temperature		117	-	137	°C
Minimum service temperature		-73.2	-	-23.2	°C
Flammability		Highly flammable			
Emissivity	*	0.81	-	0.9	

Hygro-thermal properties

Water absorption		12	-	35	%
Water vapor permeability		2.55e-13	-	1.02e-12	kg.m/s.m^2.Pa
Frost resistance		Good			

Electrical properties

Electrical conductor or insulator?		Poor insulator			
Electrical resistivity	*	2.1e14	-	7e14	µohm.cm
Dielectric constant (relative permittivity)	*	4.13	-	5.05	
Dissipation factor (dielectric loss tangent)	*	0.058	-	0.071	
Dielectric strength (dielectric breakdown)	*	1	-	2	MV/m

Optical properties

Natural color

Light brown

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Good
Sound isolation	Average

Processability (scale 1 = impractical to 5 = excellent)

Moldability	3	-	4	
Machinability	5			

Durability

Water (fresh)	Limited use
Water (salt)	Limited use
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Acceptable
UV radiation (sunlight)	Good
Wear resistance	Unacceptable
Industrial atmosphere	Limited use
Rural atmosphere	Limited use
Marine atmosphere	Limited use

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.638	-	0.703	kg/kg
Embodied energy (virgin grade production)	* 33.3	-	36.8	MJ/kg
Water Usage	* 665	-	735	l/kg

Material processing: energy

Machining, coarse, embodied energy	* 16.1	-	18.2	MJ/kg
Machining, fine, embodied energy	* 19	-	22.5	MJ/kg
Machining, grinding, embodied energy	* 21.9	-	26.8	MJ/kg

Material processing: climate change

Machining, coarse, climate change (CO2-eq)	* 1.05	-	1.19	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.24	-	1.47	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.43	-	1.75	kg/kg

Recycling and end of life

Recycle	✓			
Climate change (CO2-eq), recycling	* 0.332	-	0.366	kg/kg
Embodied energy, recycling	* 27.6	-	30.5	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 19.8	-	21.3	MJ/kg
Combustion CO2	* 1.69	-	1.78	kg/kg
Landfill	✓			
Biodegrade	✓			
A renewable resource?	✓			

Supporting information

Typical uses

Furniture, building and construction, marine and boat building, packaging, transport and vehicles, musical instruments, aircraft, modeling.

Related standards

BS 565 (1963)

Links

Producers

Reference